

Kuan-Hung Chen

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Education

University of Warwick

MSc in Statistics (with Distinction)

Focus: Statistical Inference, Data Analysis, Machine Learning, Neural Networks

Coventry, UK

Sep 2024 — Sep 2025

National Taiwan University

B.A. in Economics & B.A. in Foreign Languages and Literatures

Double Major, Overall GPA: 3.9 / 4.3. Focus: Econometrics, Game Theory

Taipei, Taiwan

Sep 2019 — Jun 2024

Experience

Market Research Intern

KANTAR Worldpanel

Taipei, Taiwan

Jun 2023 — Apr 2024

- Engineered 10+ scalable Python and VBA data pipelines to accelerate the data processing workflow, making Excel formatting and PowerPoint report production at least 20x faster.
- Analysed consumer datasets to extract actionable insights for 10+ FMCG clients, delivering reports and presentations on market trends and demand forecasts.
- Owned the end-to-end analytics lifecycle for an industry-leading client account, delivering data-driven consultancy.

Research and Projects

MSc Dissertation, University of Warwick

Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas

Sep 2025

- Engineered a vine copula-based neural network in PyTorch for mid-term probabilistic weather forecasting.
- Implemented a quasi-Bayesian recursive density estimation algorithm enabling data augmentation without retraining, improving training efficiency.
- The framework proved stable and highly accurate on chaotic time-series benchmarks and real-world WeatherBench data, demonstrating potential for practical application.

Pollparse

ALBERT-based zh-TW poll parser powering Sootow (open source)

2026

- Developed an open-source parser and training framework for Traditional Chinese poll text, tuned to the natural phrasing habits of Taiwanese users.
- Engineered a transformer fine-tuning pipeline on a pretrained ALBERT encoder for span-boundary prediction of poll titles, options, and settings.
- Designed a regex-and-lexicon rule layer converting labelled spans into structured JSON payloads, now serving as the backend API of Sootow, a live polling web app.

zh-TW NLP Product Categorizer

TF-IDF + SVC/MNB classifier served with FastAPI and Docker (open source)

2026

- Built an NLP categorization framework optimised for zh-TW product names that mix English, numerals, and Chinese characters; containerised a production-ready inference API with FastAPI and Docker.
- Achieved 70% accuracy and F1 on highly noisy, error-prone real-world data, establishing a robust performance baseline.

Intaglio

Obsidian-to-Jekyll publishing pipeline, shipped as a GitHub Action (open source)

2026

- Built a Python pipeline that converts Obsidian Markdown into Jekyll-ready output, with incremental builds and asset syncing so only changed notes are rebuilt, and hardened against non-standard real-world Markdown with unit, integration, and CLI-level tests.
- Packaged it as a reusable GitHub Action that publishes to a separate site repository through reviewable pull requests, keeping automated output auditable and reversible.

Skills

Programming: Python, SQL, C++, R, VBA, JavaScript, Ruby

Machine Learning: PyTorch, scikit-learn, NumPy, Pandas, NLP, Transformers, Time Series Forecasting

Engineering & Web: Docker, FastAPI, Git, Bash, GitHub Actions, Ruby on Rails, React, Tableau, Excel

Languages: Mandarin (Native), English (Professional; IELTS Academic 8.0), German (Intermediate)